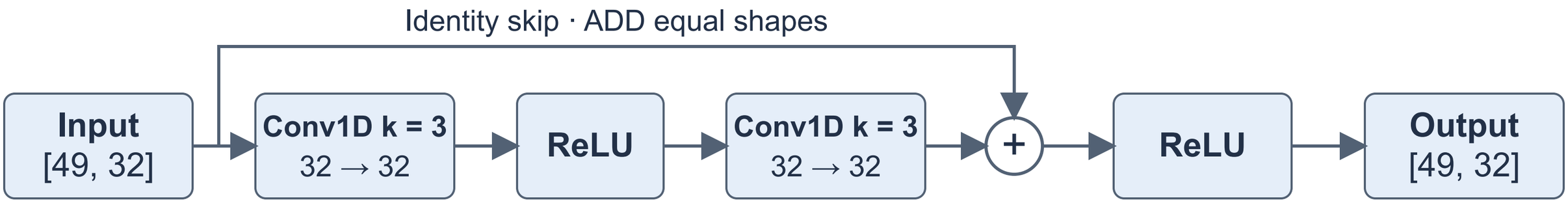
What repeats, what joins, and what dropout does

Implemented architecture · schematic teaching examples, not measured activations

A · One residual block · repeated structure ×4 per encoder



Same padding preserves 49 slots. Each block has its own weights.

Stem: Conv1D k=3 → 32 + ReLU. Four two-conv blocks give nine conv layers.

B · A trace vector is a learned summary of one entire trace

Attention receives all T rows at once and returns T updated rows.

Each row represents an entire trace. Four rows illustrate four trace summaries.

At each locus, the distance encoder reads 49 distances to named reference loci.

Ordered flatten: 49 × 64 = 3,136. Dense 32 creates one trace summary.

XYZ uses one shared centre. Training rotates all cell traces together (SO(3)).

C · CONCAT, ADD and DROPOUT have different jobs

CONCAT
[geometry 32 | metadata 16]
48 learned features
Keep entries side by side

ADD
[1, 2] + [3, 4]
= [4, 6]
Same width: 2 → 2

DROPOUT
Train: mask + rescale
Width stays 48
Prediction: pass through

Metadata: area z-score; valid traces / 4; observed fraction; area-missing flag.

Area scaling uses training cells. The other three inputs retain their defined scales.

Dropout comes before Dense 32 → ReLU → Dense 3. Softmax is used for prediction.

Observed-point masks accompany both CNNs. No coordinate imputation.

Padded traces are excluded from attention and the mean.

Geometry 32 + metadata 16 are learned features. Self-attention includes each trace itself.